IPP 1: plasma position and shape from magnetic measurements (COMPASS)

Task. From 24 magnetic measurements (16 coils, 8 flux loops) estimate 16 quantities of the plasma (axis, X point, strike points, limiter point, current, elongation, triangularity), fast enough for control. The classical reconstruction takes minutes.

What we built. A *basic network* (24 inputs, 16 outputs). The *selected model* adds 22 differences of neighbouring coils, a linear model for the plasma current and a separate network for the limiter point. For the provided noisy inputs we ship another network, trained with noise and drift.

Result. The labels are the institute's reference reconstruction, so every number is *agreement with that reconstruction*, not physical accuracy. Mean absolute error, clean inputs. *Set aside*: 30 discharges kept out of training, scored once. *5-fold*: 119 development discharges, whole discharges per fold. Latency: one core of an Apple M5, shared machine, from the 24 raw inputs.

Quantity	Selected, 30 set aside [95 % bootstrap by discharge]	Basic network, same 30	Selected, 5-fold
Axis R (mm)	0.133 [0.099, 0.177]	0.307	0.23
X point R (mm)	0.115 [0.076, 0.168]	0.161	0.211
Strike points R, 1 and 2 (mm)	0.108 [0.077, 0.144], 0.136 [0.084, 0.205]	0.212, 0.212	0.206, 0.254
Limiter point R, Z (mm)	4.55 [3.30, 6.16], 8.35 [6.81, 10.2]	10.6, 22.5	4.62, 11.8
Plasma current (kA)	0.0347 [0.0311, 0.0382]	1.20	0.0332
Latency per sample (ms)	median 0.015 (diverted), 0.027 (limited); p99 0.057, 0.093	basic network 0.0145	n/a

Sources: `results/ipp1-final.md`, `ipp1-latency-shipped.md`, `ipp1-latency.md`. The order holds in 5 of 5 seeds; the X point is only somewhat better (`ipp1-final-seeds.md`).

Other splits. Blocked split (train on the 60 lowest discharge ids, test on the 59 highest; ids look chronological, not confirmed): axis R 0.397 mm against 0.216 mm on random folds of the same 59 discharges (basic network 1.23 against 0.448 mm), with fewer training discharges (`ipp1-final.md`). Training against held-out error, axis R of the selected model: 0.095 against 0.230 mm in 5-fold (2.4 times), 0.087 against 0.133 mm on the 30 (1.5 times); basic network 1.7 and 1.2 times; the largest 5-fold ratio is 3.8 times (strike point 2 R), so it fits its training discharges better than new ones (`ipp1-traintest-gap.md`). Errors are means: the basic network's median axis R error is 0.15 mm against a mean of 0.38 mm, and limited plasmas carry 92 % of its squared axis error (`ipp1-distribution.md`).

What we did NOT show.

- Physical accuracy beyond agreement with the reference; no independent measurement was used.
- Shapes outside the data: held-out groups with the 18 unusual discharges were 1.0 to 3.0 times worse, the extreme was never held out alone (`ipp1-unusual-shapes.md`). The model gives 16 numbers, not the boundary.
- Noisy inputs for the clean model: it fails, axis R 10.15 mm against 0.38 mm (basic network, 5-fold; `ipp1-robustness-findings.md`). The noise-trained network gives 2.38 mm on the 30 (`ipp1-final.md`). The provided noisy file has white noise only; drift results are synthetic.
- Signal acquisition and a real-time system: only the network computation was timed; tails are not bounds.

Bonus analyses (one line each).

- Amount of data: 30 training discharges give axis R 1.47 mm, 119 give 0.38 mm, fixed 60 epochs; time context: three samples help on noisy inputs, hurt on clean (`ipp1-robustness-findings.md`).
- Fewer inputs: 16 coils alone 0.066, all 24 inputs 0.063, 8 flux loops alone 0.285 (summary error; `ipp1-inputs.md`).
- Stability: errors grow mostly in the last tenth of a discharge (`ipp1-time.md`).
- Disagreement of five networks follows the error, Spearman 0.46 for axis R (`ipp1-uncertainty.md`).
- Camera network: axis R 1.79 mm in colour, 1.70 mm in grey, against 0.38 mm from magnetics (`cloud-camera-main.md`).

Where everything is. Notebooks `notebooks/01` to `09`; aggregate numbers and figures in `results/`; wording rules in `docs/claims-corrections.md`. Raw data are confidential and not in the repository.